Smart Gate v3: Testing and Validation Record

1. Purpose and scope

This record holds the full test data behind Sections 2.3–2.5 of the Final Delivery Report. It covers motor-driver validation (Stages 1–2), integrated testing on the Jetson bench (Stage 3), the per-class batch detection test, and the detection-threshold trials. Tests not yet run are listed in Section 9 so the next team can pick them up.

2. Test environment

Item	Version or detail
Compute board	NVIDIA Jetson Orin Nano (OpenSI-supplied), JetPack 7.2
Inference	YOLOv5, inherited weights, 16-species class list; TensorRT 10.16.2 engine built on this board (`models/marsupial_16s.engine`)
Runtime	Docker image `smartgate` (CUDA 13.2, OpenCV with GStreamer)
Camera	Arducam 8MP V2.3 (IMX219) on CSI
Motion trigger	PIR sensor (pin 15)
Actuation	L298N driver, stepper motor (48 steps per revolution), two Hall-effect end stops
Power	6×AA pack for the motor (about 6–7 V); Jetson on its own supply
Ruleset	`config/config.json`: threshold 0.6 (0.5 inherited); OPEN = nine native species; CLOSE = Red Fox, Fallow Deer, Pig

3. Stage 1: manual motor-driver validation

Purpose. Confirm the L298N could produce both directions and a disabled state before any processor was connected.

Set-up. L298N driver, breadboard, external 9 V pack (six cells), jumper wires, red and green LEDs with current-limiting resistors standing in for the two gate directions (Figure 1). Inputs were set by hand and outputs measured with a multimeter.

Finding. The first driver gave inconsistent output on ENB and was replaced; all results below are from the replacement.

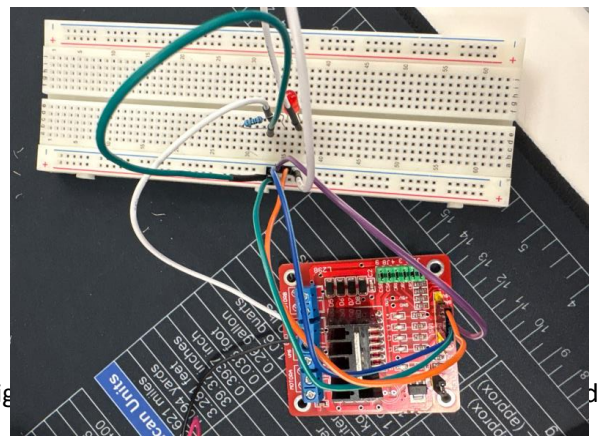


Fig d LED Validation

Test	Expected	Actual	Status
External supply	Voltage available	~9 V at the pack	Pass
Direction 1	One output active	8.17 V; first LED on	Pass
Direction 2	Polarity reversed	8.92 V; second LED on	Pass
Disabled	0 V at both outputs	~0 V at both outputs	Pass
LED alternation	LEDs alternate with direction	Alternated correctly	Pass

Outcome. The driver circuit represents both gate directions and a safe disabled state, establishing the electrical baseline for programmed control.

4. Stage 2: Raspberry Pi GPIO control

Purpose. Replace manual inputs with software-driven GPIO and confirm the same three states. A Raspberry Pi 3B+ was used because the Orin had not yet arrived; it supports Python and GPIO control, so driver logic could be validated early.

Set-up. Stage 1 circuit plus the Raspberry Pi driving IN3, IN4 and ENB from a Python script (Figure 2).

Table 1: GPIO and output voltages

Condition	IN3	IN4	ENB	Motor B terminal 1	Motor B terminal 2
Direction 1	3.29 V	0.16 V	3.34 V	0.01 V	8.89 V
Direction 2	0.16 V	3.30 V	3.35 V	8.79 V	0.11 V
Disabled	0 V	0 V	0 V	~0 V	~0 V

Table 2: LED logic sequence

Step	Expected	Actual	Status
Initial disabled state	Both LEDs off	Both LEDs off	Pass
Direction 1	Green on, red off	Green on, red off	Pass
Direction 2	Red on, green off	Red on, green off	Pass
Return to disabled	Both LEDs off	Both LEDs off	Pass

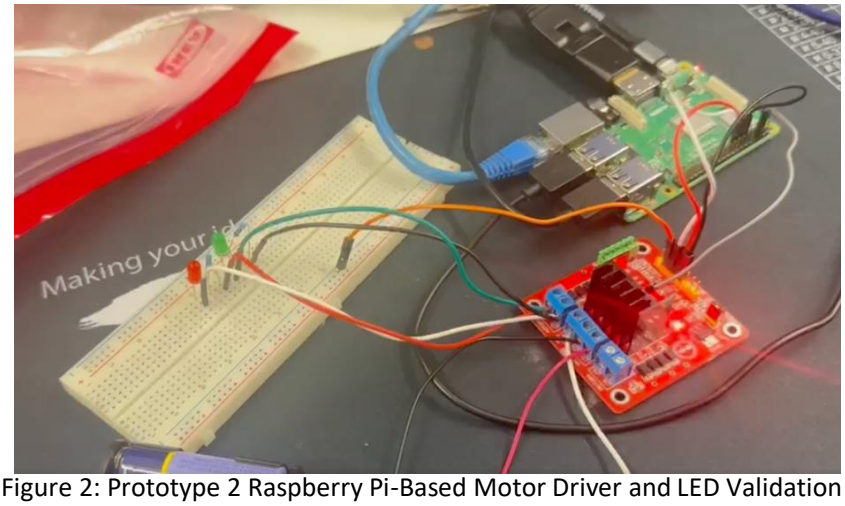


Figure 2: Prototype 2 Raspberry Pi-Based Motor Driver and LED Validation

Outcome. Programmed GPIO reproduces the manually verified directional control. Detection, decision logic and Orin compatibility were outside this stage.

5. Stage 3: integrated Jetson bench

Purpose. Test the full chain on the target hardware: PIR trigger, camera, TensorRT detection, ruleset decision, stepper motion to a Hall stop, and manual control from the dashboard.

Test	Expected	Measured	Result
Predator detection	Red Fox detected, gate closes	Fox detected at 0.69; gate closed at the close stop	Pass
Native detection	Eastern grey kangaroo detected, gate opens	Detected above the 0.6 threshold (exact confidence not recorded); gate opened at the open stop	Pass
Person in view, threshold 0.5	No gate movement	Read as Fallow Deer at 0.51–0.57; gate closed	Fail
Person in view, threshold 0.6, model only	No detection	5 of 120 frames misread (Section 7)	Partial
Person in view, threshold 0.6, integrated	No gate movement	No false close across three full cycles	Pass
Manual web control	Dashboard moves the gate	Open/Close/Stop drove the stepper to the correct stop	Pass
Stepper travel	Stops at the correct end	0.25 s per step, four-step full-step sequence, 48 steps per revolution; stops at open (pin 11) and close (pin 13)	Pass

6. Batch detection test

Purpose. Measure how reliably each class is detected, and record v3's reference accuracy.

Method. About 20 still images per class were presented to the camera at threshold 0.6. A result counts as correct when the expected class is detected above the threshold. To repeat the test without moving the gate, use `src/main/detect_check.py`, which runs the camera and detection only.

Class	Ruleset action	Images	Detected correctly	Missed	Wrong class
Red Fox	Close	20	20	0	0
Fallow Deer	Close	20	18	2	0
Pig	Close	20	17	3	0
Eastern grey kangaroo	Open	20	20	0	0
Person (not a class)	No action	20	—	—	0

Summary. Animals were detected in 75 of 80 images (94%). All five misses were predators (CLOSE classes: 55 of 60, 92%); every native image was detected (20 of 20). No person was misread in still images.

7. Detection threshold trials

Purpose. Decide whether to keep the inherited 0.5 threshold after a person was read as a Fallow Deer.

Threshold	Mode	Cycle	Frames with person	Frames misread	Gate closed?
0.5	Integrated	1	Not recorded	Read as Fallow Deer at 0.51–0.57	Yes
0.6	Model only (no actuation)	1	40	0	n/a

Threshold	Mode	Cycle	Frames with person	Frames misread	Gate closed?
0.6	Model only (no actuation)	2	40	3	n/a
0.6	Model only (no actuation)	3	40	2	n/a
0.6	Integrated	3 full cycles	Not recorded	—	No

Result. Raising the threshold to 0.6 cut model-only misreads to 5 of 120 frames (about 4%) and removed false closes in integrated runs, while the fox was still detected at 0.69. The fox's margin over 0.6 is about 0.09.

8. Comparative design decisions

Choice	Options compared	Evidence	Decision
Hall sensor supply	3.3 V vs 5 V	Sensor needs about 4.5 V; at 5 V the output stays at a safe 3.34 V	5 V
Gate demonstration	LEDs vs stepper with Hall stops	LEDs show direction only; the stepper shows motion and end positions	Stepper
Detection threshold	0.5 vs 0.6	Section 7	0.6

9. Not yet tested

- **Inference latency:** time from frame capture to detection result, as a mean and range over at least 50 frames.
- **Sustained run:** operation over an agreed period without manual recovery.
- **More classes:** further native species, and live animals or video rather than still images.
- **PIR behaviour:** response to a still animal and confirmation of pin roles.
- **Multi-frame decision:** effect of requiring several agreeing frames before the gate moves.

10. Figures

- Figure 1: Stage 1 breadboard set-up (`docs/testing/fig1_stage1_breadboard.jpg`)
- Figure 2: Stage 2 Raspberry Pi set-up (`docs/testing/fig2_stage2_raspberry_pi.jpg`)
- Figure 3: integrated Jetson bench during Stage 3 (`docs/testing/fig3_stage3_bench.jpg`)